Ignacio de la Serna

publishes as Ignacio Serna

INDEPENDENT AI RESEARCHER

+34 646 90 94 94 · serna@mpib-berlin.mpg.de · ignacioserna.github.io · Google Scholar

Last updated October 2026

EDUCATION

California Institute of Technology — Pasadena, CA, U.S.A.
Visiting Student (Ph.D.) Researcher — 2022 – 2023
Advisor: Pietro Perona

Autonomous University of Madrid — Madrid, Spain
Doctor of Philosophy in Electrical Engineering and Computer Science (Ph.D.) — 2020 – 2024
Advisors: Aythami Morales (primary) and Javier Ortega-Garcia (secondary)
Dissertation: "Understanding Bias at Multiple Levels in Computer Vision Algorithms with a focus on Face Recognition"

National University of Distance Education — Madrid, Spain
Master of Science in Artificial Intelligence — 2018 – 2020

Autonomous University of Madrid — Madrid, Spain
Bachelor of Science in Mathematics — 2012 – 2018
Bachelor of Science in Computer Science — 2012 – 2018

EXPERIENCE — ACADEMIA

University of Oxford — Oxford, UK
Visiting Researcher, Technical Safety & Governance (TSG) Lab — 2026.09 – present

IMDEA Software Institute — Madrid, Spain
Visiting Faculty — 2026.03 – present

IE University — Madrid, Spain
Adjunct Professor, School of Science and Technology — 2026.01 – present

Max Planck Institute for Human Development — Berlin, Germany
Visiting Researcher, Center for Humans and Machines — 2026.02 – present
Post-doctoral Researcher, Center for Humans and Machines — 2024.02 – 2026.01

Autonomous University of Madrid — Madrid, Spain
Research Assistant, Biometrics and Data Pattern Analytics (BiDA) Lab — 2019.02 – 2020.08
Research Intern, Biometric Recognition Group – ATVS — 2015.09 – 2016.05

EXPERIENCE — INDUSTRY

GMV Aerospace and Defence S.A. — Madrid, Spain
GNSS (Global Navigation Satellite Systems) Junior Engineer — 2018.02 – 2018.09
Research Trainee — 2017.10 – 2018.02

HONORS & DISTINCTIONS

2026 Open Philanthropy / Coefficient Giving Grant for **Technical AI Safety Research** ($155,000)

2020 FPI (pre-doctoral Research Staff Training) scholarship, Autonomous University of Madrid

TALKS & SEMINARS

Invited Talk — IMDEA Software Institute
Madrid Institute for Advanced Studies | Madrid, Spain | January 2026
"Bias in Deep Face Models"

Short Course — Master's Program in Artificial Intelligence
Technical University of Madrid, Spain | June 2025
"Machine Behavior and Machine Culture"

Invited Talk — Fairness in Biometric Systems Workshop
Intl. Conf. on Pattern Recognition (ICPR) | Montreal, Canada | August 2022
"Where are we on measuring bias?"

Tutorial — EU H2020 Secure & Privacy-preserving Biometrics Summer School
Autonomous University of Madrid, Spain | July 2022
"Hands-on: NLP Tools"

WORK UNDER REVIEW

1. H. Yakura, E. Lopez-Lopez, L. Brinkmann, **I. Serna**, P. Gupta, and I. Rahwan. "Empirical evidence of Large Language Models' influence on human spoken communication". Under review, Nature.

PUBLICATIONS

REFEREED JOURNAL & CONFERENCE PAPERS (Google Scholar)

1. **I. Serna** (Accepted). "Discovering Intersectional Bias via Directional Alignment in Face Recognition Embeddings". IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027, Orlando, FL.
2. **I. Serna**, A. Morales, and J. Fierrez. "Unraveling Machine Behavior by Multi-Level Bias Analysis and Detection: Methodology and Application to Computer Vision", Pattern Recognition (accepted).
3. H. Yakura, E. Lopez-Lopez, L. Brinkmann, **I. Serna**, P. Gupta, I. Soraperra, T. F. Eisenmann, I. Rahwan (April 2026). "Empirical Evidence of a Bi-Directional Linguistic Interaction between Humans and Machines", Intl. Conf. on the Evolution of Language (EVOLANG), Plovdiv, Bulgaria.
4. A. H. Artiles, H. Yakura, L. Brinkmann, M. C. Sola, H. A. Alhaija, **I. Serna**, N. Rahaman, B. Schölkopf, I. Rahwan (Dec 2025). "Cultural Alien Sampler: Open-ended Art Generation Balancing Originality and Coherence", Advances in Neural Information Processing Systems (NeurIPS), Creative AI Track, San Diego, CA, USA.
5. M. Knott, **I. Serna**, E. Mann and P. Perona (Feb 2025). "A Rapid Test for Accuracy and Bias of Face Recognition Technology", IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV), Tucson, AZ, USA.
6. A. Hernandez, L. Brinkmann, **I. Serna**, N. Rahaman, H. A. Alhaija, H. Yakura, M. C. Sola, B. Schölkopf and I. Rahwan (Dec 2024). "Alien Recombination: Exploring Concept Blends Beyond Human Cognitive Availability in Visual Art", Workshop on Creativity & Generative AI, Neural Information Processing Systems (NeurIPS), Montreal, Canada.
7. A. Peña, A. Morales, J. Fierrez, **I. Serna**, J. Ortega-Garcia, I. Puente, J. Cordova and G. Cordova (Aug 2023). "Leveraging Large Language Models for Topic Classification in the Domain of Public Affairs", IAPR Intl. Conf. on Document Analysis and Recognition (ICDAR) Workshops, San José, CA, USA.
8. A. Peña, **I. Serna**, A. Morales, J. Fierrez, A. Ortega, A. Herrarte, M. Alcantara & J. Ortega-Garcia (Jun 2023). "Human-Centric Multimodal Machine Learning: Recent Advances and Testbed on AI-Based Recruitment", SN Computer Science, vol. 4, no. 5, p. 434.
9. M. Ghafourian, J. Fierrez, R. Vera-Rodriguez, A. Morales and **I. Serna** (June 2023). "OTB-morph: One-time Biometrics via Morphing", Machine Intelligence Research, vol. 20, no. 6, pp. 855–871. ISSN: 2731-5398.
10. D. DeAlcala, **I. Serna**, A. Morales, J. Fierrez, and J. Ortega-Garcia (June 2023). "Measuring Bias in AI Models: A Statistical Approach Introducing N-Sigma", IEEE Conf. on Computers, Software, and Applications (COMPSAC), Torino, Italy.
11. **I. Serna**, A. Morales, J. Fierrez and N. Obradovich (April 2022). "SensitiveLoss: Improving Accuracy and Fairness of Face Representations with Discrimination-Aware Deep Learning", Artificial Intelligence, vol. 305.
12. A. Morales, J. Fierrez, A. Acien, R. Tolosana and **I. Serna** (Feb 2022). "SetMargin Loss Applied to Deep Keystroke Biometrics with Circle Packing Interpretation", Pattern Recognition, vol. 122.

13. **I. Serna**, D. DeAlcala, A. Morales, J. Fierrez, J. Ortega-Garcia (Feb 2022). "IFBiD: Inference-Free Bias Detection", AAAI Workshop on Artificial Intelligence Safety (SafeAI), Vancouver, Canada.
14. M. Ghafourian, J. Fierrez, R. Vera-Rodriguez, **I. Serna** and A. Morales (Jan 2022). "OTB-morph: One-Time Biometrics via Morphing Applied to Face Templates", IEEE/CVF Winter Conf. on Applications of Computer Vision Workshops (WACVw), Waikoloa, HI, USA.
15. J. Hernandez-Ortega, J. Fierrez, **I. Serna** and A. Morales (Dec 2021). "FaceQgen: Semi-Supervised Deep Learning for Face Image Quality Assessment", IEEE Intl. Conf. on Automatic Face and Gesture Recognition (FG), Jodhpur, India.
16. A. Peña, **I. Serna**, A. Morales, J. Fierrez and A. Lapedriza (Sep 2021). "Facial Expressions as a Vulnerability in Face Recognition", IEEE Intl. Conf. on Image Processing (ICIP), Anchorage, AK, USA, pp. 2988–2992.
17. **I. Serna**, A. Peña, A. Morales and J. Fierrez (Jan 2021). "InsideBias: Measuring Bias in Deep Networks and Application to Face Gender Biometrics", IAPR Intl. Conf. on Pattern Recognition (ICPR), Milan, Italy.
18. A. Peña, **I. Serna**, A. Morales and J. Fierrez (Oct 2020). "FairCVtest Demo: Understanding Bias in Multimodal Learning with a Testbed in Fair Automatic Recruitment", ACM Intl. Conf. on Multimodal Interaction (ICMI), Utrecht, The Netherlands (also presented at ECCV 2020).
19. A. Peña, **I. Serna**, A. Morales and J. Fierrez (June 2020). "Bias in Multimodal AI: Testbed for Fair Automatic Recruitment", IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops (CVPRw), Seattle, USA (also presented at ICML 2020 Workshop on Human-in-the-Loop Learning).
20. **I. Serna**, A. Morales, J. Fierrez, M. Cebrian, N. Obradovich and I. Rahwan (Feb 2020). "Algorithmic Discrimination: Formulation and Exploration in Deep Learning-based Face Biometrics", AAAI Workshop on Artificial Intelligence Safety (SafeAI), New York, USA.

CONFERENCE ORGANIZING COMMITTEE

- Member of Local Organizing Committee, IEEE/IAPR International Conference on Biometrics (ICB), 2013.
- Member of Local Organizing Committee, IAPR Iberian Conference on Pattern Recognition and Image Analysis (IbPRIA), 2019.

REVIEWER

Journals IEEE Transactions on Image Processing · IEEE Open Journal of Signal Processing · IEEE Transactions on Information Forensics & Security · IEEE Transactions on Biometrics, Behavior, and Identity Science · Applied Soft Computing (Elsevier) · Collective Intelligence (Sage / ACM) · The Computer Journal (BCS) · PLOS ONE · Nature Communications Psychology / Scientific Reports · AI & SOCIETY (Springer Nature)

Conferences CVPR · ECCV · NeurIPS · ICLR · WACV · International Joint Conference on Biometrics (IJCB)

PROJECT PARTICIPATION

COMPETITIVE PUBLIC PROJECTS

HumanCAIC — Enhanced Behavioral Biometrics for Human-Centric AI in Context (TED2021-131787B-I00)
Sep 2022 – Dec 2024

European Union – NextGenerationEU Funds · €314,000 · PIs: Julián Fierrez and Aythami Morales
Development and study of new human-centered learning strategies.

BBforTAI — Biometrics and Behavior for Unbiased and Trustworthy AI with Applications (PID2021-127641OB-I00)
Jan 2022 – Dec 2024

Spanish Ministry of Science and Innovation · €273,000 · PIs: Aythami Morales and Julián Fierrez
New reliable machine learning technologies based on biometric and behavioral modeling.

BIBECA — Biometric Recognition and Behavior Analysis for Safe, Context-Sensitive Human-Machine Interaction (RTI2018-101248-B-I00R)
Jan 2019 – Dec 2021

Spanish Ministry of Science, Innovation and Universities · €240,000 · PIs: Aythami Morales and Julián Fierrez
New biometric recognition technologies based on behavior.

TECHNOLOGY TRANSFER PROJECTS TO PRIVATE SECTOR

Detecting Fraud in Facial Biometric Systems Mar 2022 – Nov 2022
VERIDAS · €48,200 · PI: Aythami Morales Moreno
Development of attack-detection modules in facial recognition systems.

VINCESAI-ARGOS — Development of a Politics Intelligence Platform Oct 2021 – Nov 2022
Vinces Consulting · €25,380 · PI: Aythami Morales Moreno
Platform for information-source analysis and massive text processing.

CECA-Analytics Apr 2014 – May 2016
Cecabank S.A. · confidential (>€200,000) · PI: Javier Ortega-García

TEACHING

IE University Madrid, Spain
CSAI.2.M.B — AI: Machine Learning Foundations Spring 2026
Instructor · Enrollment: 40

Autonomous University of Madrid Madrid, Spain
EE446 — Signal Analysis and Information Modeling (Undergraduate) Spring 2022
Teaching Assistant · Enrollment: 40 · Co-instructor: Javier Ortega-García
EE52 — Biometric Recognition (Master's) Spring 2022
Teaching Assistant · Enrollment: 20 · Co-instructor: Javier Ortega-García
University Program for Seniors Nov 2023
Instructor · Enrollment: 50 · Co-instructor: Aythami Morales

ADVISEES

MASTER'S STUDENTS

2023 – 2024 Alejandro Hernandez Artiles (CS, Technische Universität München)
2022 – 2023 Sofía Bosch Pastor (EE, Autonomous University of Madrid)
2020 – 2021 Alejandro Romero del Campo (EE, Autonomous University of Madrid)

UNDERGRADUATE STUDENTS

2022 – 2023 Javier Culebras Martínez (CS, Autonomous University of Madrid)
2021 – 2022 Rosa Hornero Romera (EE, Autonomous University of Madrid)

ADDITIONAL INFORMATION

Languages: Spanish, English, Italian.